

Geometrically Inductive Self-Supervised Learning for EEG

A Design Study of Spatial Attention Across Sessions, Subjects, and Montages

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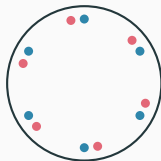
ECEN 525 – Brain–Computer Interaction

Santa Clara University

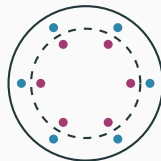
Problem: EEG is non-stationary – by default

EEG drifts on **three geometric axes**:

- **Session** — cap shifts day to day.
- **Subject** — head sizes differ; C3 is not the same point.
- **Montage** — 2 to 256 electrodes in different layouts.



Session
day 1 vs day 2



Subject
small vs large head



Montage
64 ch vs 3 ch

The geometry has changed — per-electrode parameters cannot

Codex models (LaBraM, EEGPT):

identity-indexed embeddings.

Channel-independent (BIOT): no spatial structure at all.

Neither uses scalp geometry as an *input*.

Solution: spatial attention conditioned on g_{ij}

Geometric descriptor.

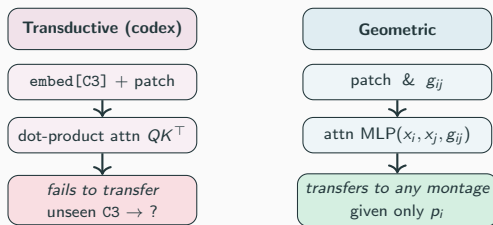
$$g_{ij} = [\|p_i - p_j\|, p_i - p_j] \in \mathbb{R}^4$$

Distance + signed displacement. No per-electrode parameters; same function for any montage.

Three options for where g_{ij} enters:

- **G1** (score) – “which electrodes attend”
- **G2** (value) – “what they exchange”
- **G3** (both) – combined

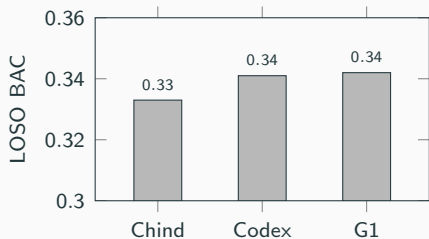
Training: BYOL-style momentum alignment + masked-patch reconstruction



raw EEG → patch → spatial attn → temporal → momentum target (EMA)

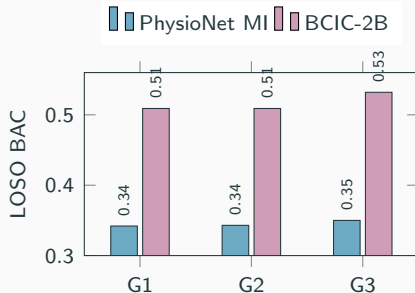
In-distribution: geometric = codex > channel-independent. G3 wins.

E1 – PhysioNet MI LOSO (n=105)



Wilcoxon (G1 vs codex): $p = 0.911$

E5 – G1 / G2 / G3 ablation



G3 (geometry at *both* score and value) wins on both datasets.

- In-distribution: geometric and codex tie; both beat channel-independent.
- Within geometric: G3 dominates G1/G2 – geometry contributes a small signal at each slot it enters.

Cross-montage transfer (64 ch $\rightarrow$ 3 ch): codex-nn matches geometric



BCIC-2B LOSO BAC (n=9, zero-shot)

Chance = 0.50 (binary). Dashed line = chance.

The proposal's headline test:

*Pretrain on PhysioNet MI (64 ch);
zero-shot transfer to BCIC-2B (3 ch).*

Codex + nearest-neighbour fallback
(0.525) slightly **beats** the geometric
encoder (0.509).

The nn fallback uses 3-D coordinates to
assign each unseen electrode to its
closest pretraining entry – it exploits
the same geometry at transfer time,
without learning a geometric attention
bias.

The predicted ordering inverts.

Conclusion: an honest read, a sharper target

What we learned:

- Spatial structure helps; geometric = codex at pilot scale.
- **G3** is the strongest geometric configuration.
- **Codex + nn** is a tougher transfer baseline than the proposal anticipated.

Three concrete next steps:

1. Push G3 with **richer descriptors** (spherical, geodesic).
2. **Scale up** pretraining and re-run E2c.
3. Test on **high-density montages** where codex-nn breaks down.

Predicted vs. observed		
	Predicted	Observed
In-distribution	tie	tie
Cross-subject	geometric +	tie
Cross-montage	geometric ++	codex-nn $\approx$ +

[github.com/tianxin-scu/
geometric-eeg-ssl](https://github.com/tianxin-scu/geometric-eeg-ssl)

Key References

- Wang et al. **EEGPT**. NeurIPS 2024 – training recipe (momentum + reconstruction).
- Grill et al. **BYOL**. NeurIPS 2020 – momentum-encoder alignment.
- He et al. **MAE**. CVPR 2022 – masked-patch reconstruction.
- Hamilton, Ying, Leskovec. **GraphSAGE**. NeurIPS 2017 – inductive vs. transductive vocabulary.
- Veličković et al. **GAT**. ICLR 2018 – attention over relational features.
- Jiang et al. **LaBraM**. ICLR 2024; Yang et al. **BIOT**. NeurIPS 2023 – codex / channel-independent baselines.
- Schalk et al. 2004 – PhysioNet MI; Tangermann et al. 2012 – BCIC-2B; Kemp et al. 2000 – Sleep-EDF.

Full reference list and per-experiment details in the project report.